parse_segments_data.py

Goal: create strava_segments.db from training.db
Working databse: training.db. table name: segments_data
New database: strava_segments.db

For a given activity_id, and the corresponding athlete_id, date, and tour_year there a segment data.
The segment column in training.db is a JSON BLOB which looks like

For example: for activity_id 6929407944 and athlete_id 4018952 the blob looks like

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{"segment_name": ["Ploubezre-K\u00e9rauzern",… , "Sprint de cinq croix"],

"segment_time": ["9:20", "9:24", …, "51s"],

"segment_speed": ["27.6km/h", "25.1km/h", …, "31.4km/h"],

"watt": ["177W", "200W", …, "179W"],

"heart_rate": ["\u2014", "\u2014", …, "\u2014"],

"segment_distance": ["4.29km", "3.93km", …, "0.44km"],

"segment_vert": ["15m", "30m", …, "3m"],

"segment_grade": ["0.2%", "0.8%", …, "0.5%"],

"VAM": ["", "", …, ""]}

The goal is to parse the BLOB into columns with all athlete-dependent measured properties of an activity into a forming the table. Note that (activity_id, athlete_id, segment_name) can't be not keys because they are not unique so we will add a row-counter (id) as a primary key

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```
CREATE TABLE segment_activity (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    activity_id INTEGER,
    athlete_id INTEGER,
    segment_name TEXT,
    segment_distance_km REAL,
    segment_vert_m REAL,
    segment_grade REAL,
    segment_time_seconds INTEGER,
    segment_speed_kph REAL,
    power_watt INTEGER,
    heart_rate_bpm INTEGER,
    vertical_ascent_m REAL,
```

```
    FOREIGN KEY (athlete_id) REFERENCES segments_data(athlete_id),
    FOREIGN KEY (segment_name) REFERENCES segments_data(segment_name)
)
```

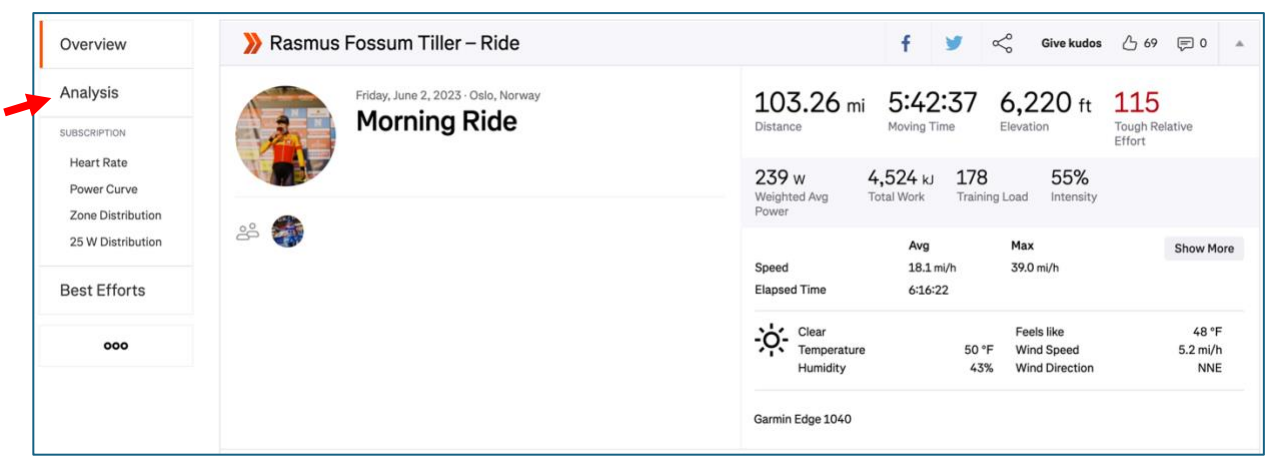
The table now looks like:

id	activity_id	athlete_id	segment_name	segment_distance_km	segment_vert_m	segment_grade	segment_time_seconds	segment_speed_kph	power_watt	heart_rate_bpm	vertical_ascent_m
1	6929407944	4018952	Ploubezre-Kérauzern	4.29	15.0	0.2	560	27.6	177	NULL	NULL
2	6929407944	4018952	Kerauzern-Plouaret	3.93	30.0	0.8	564	25.1	200	NULL	NULL
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

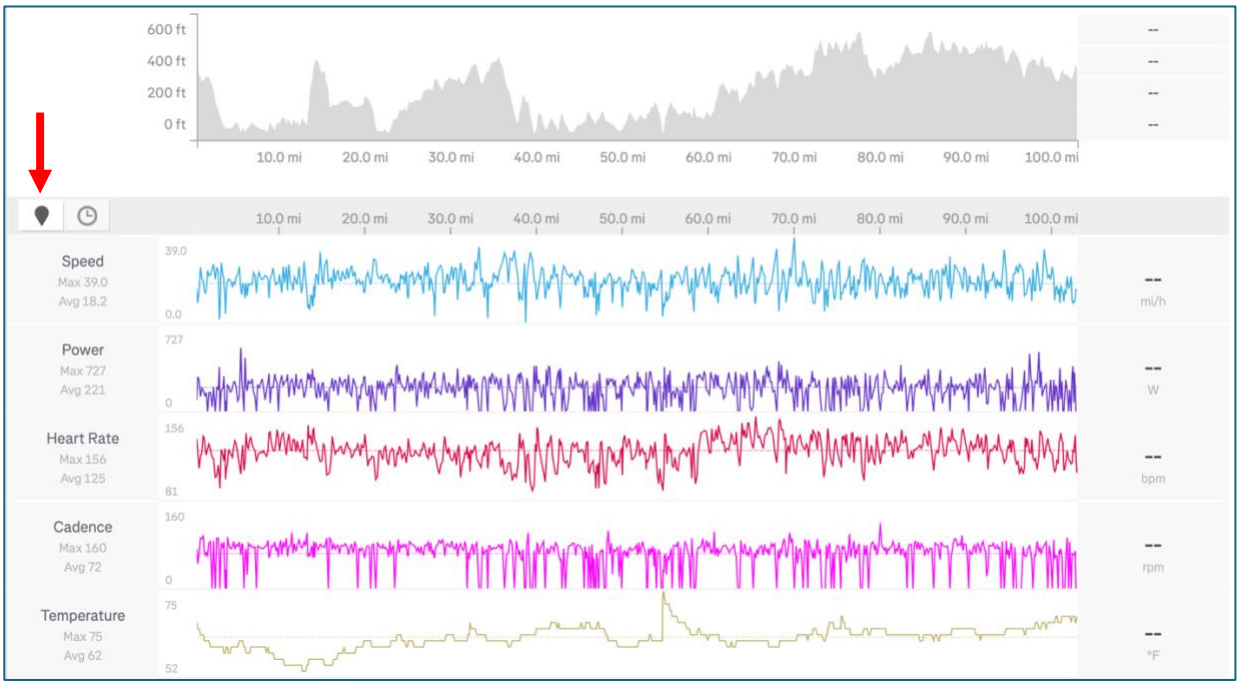
Challenges:

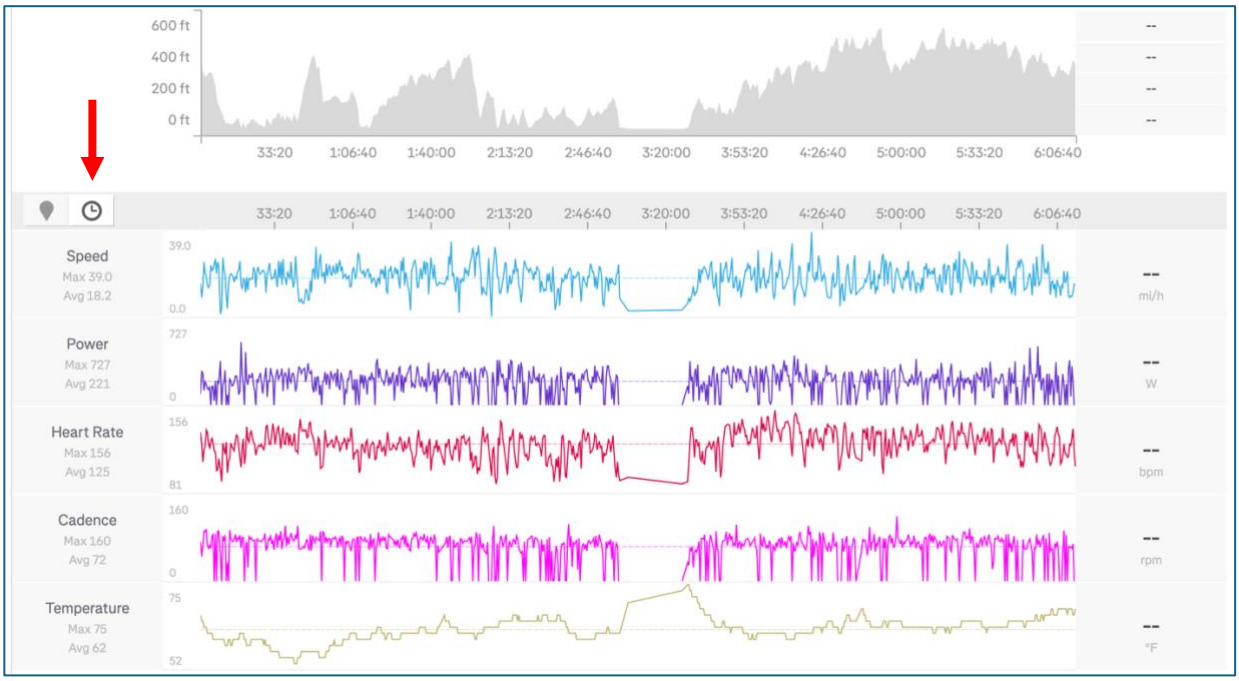
1. A training activity is broken into an arbitrary number of segments chosen by the rider.
2. A training activity could have repeated segments and circuits. Even if we get a unique segment_id, if a rider goes over that segment multiple times, then there is no way to distinguish one segment information from another. Thus, if a rider goes out and rides a circuit course, the (activity_id, athlete_id, segment_id) would all be the same

We can avoid this issue by getting the “Analysis” data of a given Strava activity.
For example, consider the for activity for athlete_id 3905720 and activity_id 9188311589.
Note: depending on location the units are either imperial or metric.



Clicking on the Analysis tab on the left we can see two types of data with a starting the zero-point. The first tab shows the distance data for that activity and the other tab shows time data. **Is there a way to scrape this data?**





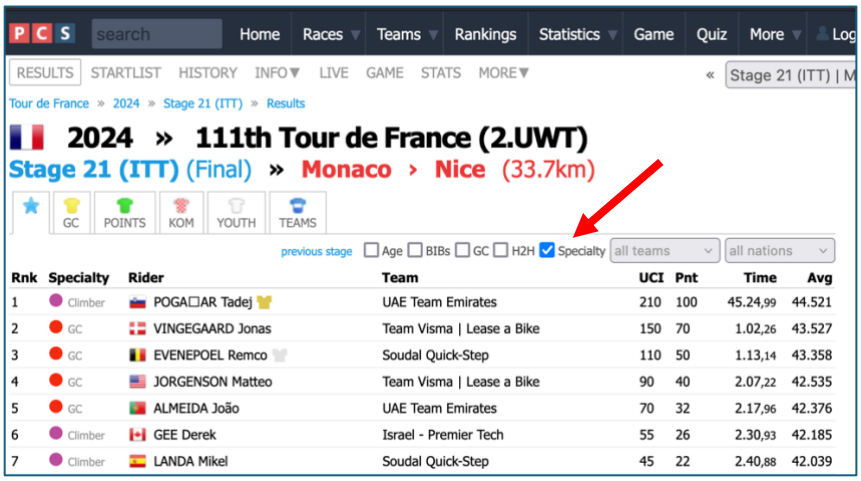
But all this depends on our goal. So here are some of our ideas for project goal

1. Classify any segment of road (length above between 1 to 5 km say) based on difficulty. Either on a scale of 1-10 or simply an (easy, medium, hard) difficulty ranking. We know the elevation and grade (slope) the segments posted by Strava. We can then construct a “normalized” representative segment for different kinds of terrain (flat, hilly, mountainous). Thus, for the actual tour we can break up the course (stage) based on difficulty ranking
    - Problem: there is no natural break in a particular segment

2. Classify a biker using the tour’s official classification (Climber, GC, Classic, Time-Trial Specialist, Hills, Sprint) . This might require us to scrape data of non-pro athletes to have a background. Create a profile of

a rider on several different axes for example (speed, strength, endurance) and predict how they are going to do in a specific stage.

- We can test our classification against the procyclingstats.com grouping.



Rnk	Specialty	Rider	Team	UCI	Pnt	Time	Avg
1	Climber	POGA□AR Tadej	UAE Team Emirates	210	100	45.24,99	44.521
2	GC	VINGEGAARD Jonas	Team Visma | Lease a Bike	150	70	1.02,26	43.527
3	GC	EVENEPOEL Remco	Soudal Quick-Step	110	50	1.13,14	43.358
4	GC	JORGENSON Matteo	Team Visma | Lease a Bike	90	40	2.07,22	42.535
5	GC	ALMEIDA João	UAE Team Emirates	70	32	2.17,96	42.376
6	Climber	GEE Derek	Israel - Premier Tech	55	26	2.30,93	42.185
7	Climber	LANDA Mikel	Soudal Quick-Step	45	22	2.40,88	42.039

3. Predict competition among elite riders. Use the split-time to rank the riders. As with most racing, if a rider is taking longer than their previous trial (or previous k trials) on a 'similar' segment, then we denote them as having a positive split time, say +0.8, and give them an indication of red. Conversely, we give them a negative split time, -1.2, and show them in green if they are taking less time to complete a 'similar' segment. I suspect for time-trial stages the units are in seconds and for regular stages the units are minutes.
    - Add interactive dashboard?

4. Based on the finishing times can we say which of the three tours is the most competitive? In addition, are there any dominant teams/riders in the tours?

5. Given the data from a certain training activity, is it possible to say how the activity would compare to a tdf activity?

Questions/remarks about data:

1. Do we need to/hope to Scrape segements_data and stats_data for Vuelta?
2. Find missing segments or stats data and scrape it (if it is from the first 10 riders of a stage, can we find it somewhere else?)
3. Figure out how much (percentage) of the top 100 finishers are represented in the Strava table
4. Can we find the official tdf or giro segments (if it exists) for a given year (as gps coordinates of the start/stop locations of the stages?)
5. Would scarping the ‘overview” tab of an activity tell us anything new?